

Car Diagnostics API 2.0.0 OAS 3.1

[/openapi.json](#)

API for emissions compliance and engine health analysis.

default



GET

/ Root

⌵

Emissions Compliance



GET

/emissions/data Fetch Emissions Data

⌵

Fetch emissions data from the database.

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/emissions/data' \
  -H 'accept: application/json'
```

⌵

Request URL

```
http://127.0.0.1:8000/emissions/data
```

Server response

Code	Details
200	Response body ⌵

Code

```
"control_module_voltage",
"absolute_load",
"commanded_equiv_ratio",
"relative_throttle_pos",
"ambient_air_temp",
"throttle_pos_b",
"accelerator_pos_d",
"accelerator_pos_e",
"throttle_actuator",
"fuel_type",
"long_o2_trim_b1",
"unknown"
],
"rows": [
  {
    "timestamp_obd": "1734340661",
    "status": "<obd.OBDResponse.Status object at 0x7d80417f9160>",
    "fuel_status": "('Closed loop, using oxygen sensor feedback to determine fuel mix', '')",
    "engine_load": "23.1372549",
    "coolant_temp": "97",
    "short_fuel_trim_1": "-1.5625",
    "long_fuel_trim_1": "0.78125",
    "intake_pressure": "36",
    "rpm": "1173.5",
    "speed": "26",
    "timing_advance": "15"
```



Download

Response headers

```
content-length: 328692
content-type: application/json
date: Sat,11 Jan 2025 15:25:44 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links
Media type application/json Controls Accept header. Example Value Schema "string"		

GET

/emissions/preprocess Preprocess Emissions Data



Preprocess emissions data and log the results.

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/emissions/preprocess' \
  -H 'accept: application/json'
```



Request URL

```
http://127.0.0.1:8000/emissions/preprocess
```

Server response

Code

Details

200

Response body

```
{
  "message": "Data preprocessing successful."
}
```



Download

Response headers

```
content-length: 44
content-type: application/json
date: Sat, 11 Jan 2025 15:25:59 GMT
server: uvicorn
```

Responses

Code

Description

Links

200

Successful Response

No links

Media type

application/json



Controls Accept header.

Example Value | Schema

```
"string"
```

POST

/emissions/train Train Emissions Model



Train the emissions model.

Parameters

[Cancel](#)

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/emissions/train' \
  -H 'accept: application/json' \
  -d ''
```



Request URL

<http://127.0.0.1:8000/emissions/train>

Server response

Code

Details

200

Response body

```
{
  "message": "Model training successful."
}
```



Download

Response headers

```
content-length: 40
content-type: application/json
date: Sat, 11 Jan 2025 15:27:01 GMT
server: uvicorn
```

Responses

Code

Description

Links

200

Successful Response

No links

Media type

application/json



Controls Accept header.

Example Value | Schema

```
"string"
```

GET

/emissions/results Get Model Results



Get test results and serve the confusion matrix as an image.

Parameters

[Cancel](#)

No parameters

[Execute](#)[Clear](#)

Responses

Curl

```
curl -X 'GET' \  
  'http://127.0.0.1:8000/emissions/results' \  
  -H 'accept: application/json'
```



Request URL

http://127.0.0.1:8000/emissions/results

Server response

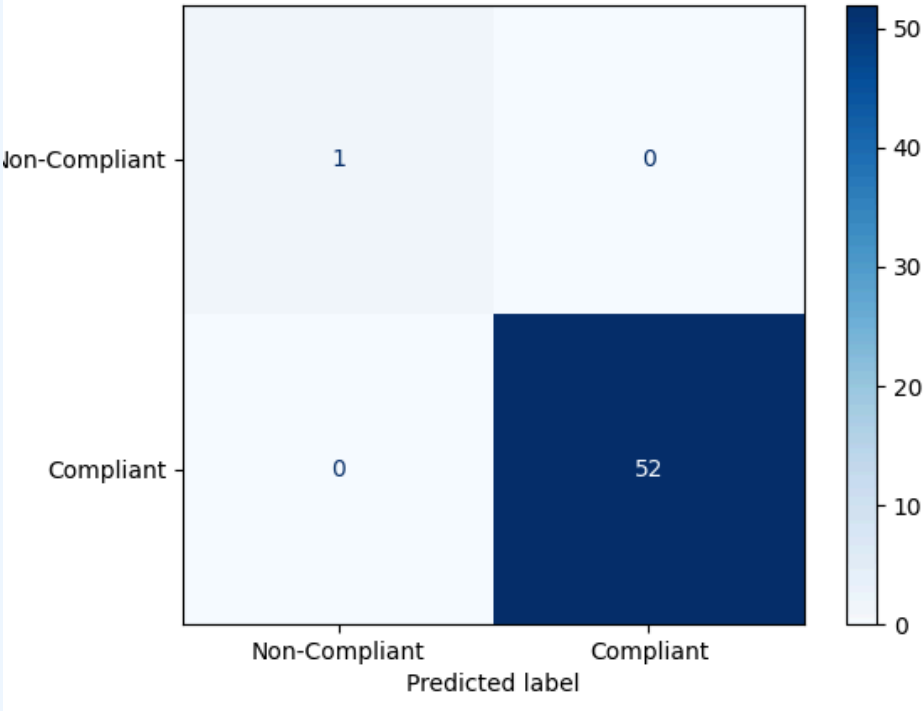
Code**Details**

200

Response body

Code

Details



Response headers

```
content-type: image/png
date: Sat,11 Jan 2025 15:27:23 GMT
server: uvicorn
transfer-encoding: chunked
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

"string"

Engine Health

GET

/engine-health/data

Fetch Engine Health Data

Fetch engine health data from the database.

Parameters

[Try it out](#)

No parameters

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/engine-health/data' \
  -H 'accept: application/json'
```



Request URL

```
http://127.0.0.1:8000/engine-health/data
```

Server response

Code

Details

200

Response body

```
{
  "columns": [
    "timestamp_obd",
    "status",
    "fuel_status",
    "engine_load",
    "coolant_temp",
    "short_fuel_trim_1",
    "long_fuel_trim_1",
    "intake_pressure",
    "rpm",
    "speed",
    "timing_advance",
    "intake_temp",
    "throttle_pos",
    "o2_sensors",
    "o2_b1s2",
    "obd_compliance",
    "run_time",
    "pids_b",
    "distance_w_mil",
    "evaporative_purge",
    "fuel_level",
    "warmups_since_dtc_clear",
    "distance_since_dtc_clear",
    "barometric_pressure",
    "o2_s1_wr_current",
    "catalyst_temp_b1s1",
  ]
}
```

[Download](#)

Response headers

```
content-length: 328692
content-type: application/json
date: Sat,11 Jan 2025 15:24:37 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links
Media type application/json Controls Accept header. Example Value Schema "string"		

GET

/engine-health/preprocess

Preprocess Engine Health Data

Preprocess engine health data and log the results.

Parameters

Try it out

No parameters

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/engine-health/preprocess' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/engine-health/preprocess

Server response

Code	Details
200	Response body <pre>{ "message": "Data preprocessing successful." }</pre> Download Response headers <pre>content-length: 44 content-type: application/json date: Sat, 11 Jan 2025 15:24:48 GMT</pre>

Code

Details

server: uvicorn

Responses

Code

Description

Links

200

Successful Response

No links

Media type

application/json

Controls Accept header.

Example Value | Schema

"string"

POST

/engine-health/train

Train Engine Health Model

Train the engine health model.

Parameters

Try it out

No parameters

Responses

Curl

curl -X 'POST' \
'http://127.0.0.1:8000/engine-health/train' \
-H 'accept: application/json' \
-d ''

Request URL

http://127.0.0.1:8000/engine-health/train

Server response

Code

Details

200

Response body

{
 "message": "Model training successful.",
 "confusion_matrix": [

Code

Details

```
53,
0
],
[
0,
0
]
}
```



Download

Response headers

```
content-length: 74
content-type: application/json
date: Sat,11 Jan 2025 15:24:54 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

"string"

GET

/engine-health/visualize

Visualize Engine Health Results

^

Visualize engine health model results.

Parameters

Try it out

No parameters

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/engine-health/visualize' \
-H 'accept: application/json'
```



Request URL

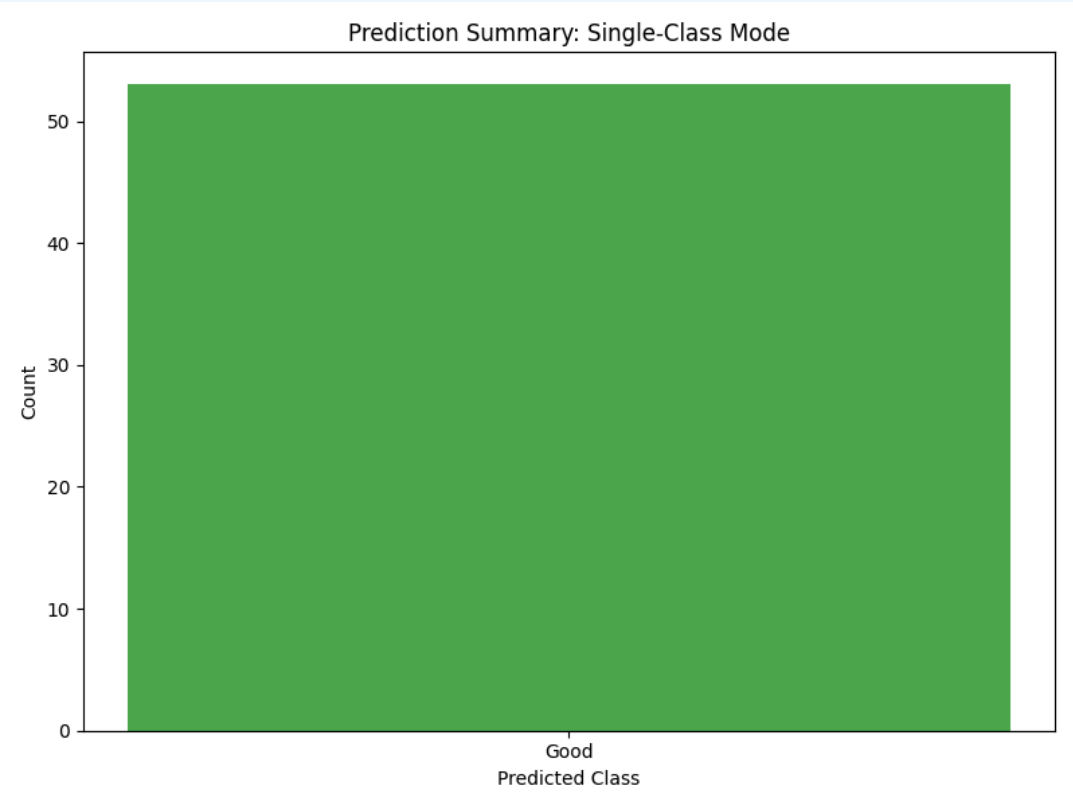
http://127.0.0.1:8000/engine-health/visualize

Server response

Code	Details
------	---------

200

Response body



Response headers

```
content-type: image/png
date: Sat,11 Jan 2025 15:25:06 GMT
server: uvicorn
transfer-encoding: chunked
```

Responses

Code	Description	Links
------	-------------	-------

200	Successful Response	No links
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Media type

application/json ▼

Controls Accept header.

Example Value | Schema

```
"string"
```